

Michael Keohane

Full-stack software engineer combining zero-to-one product execution, large-scale systems, and production LLM engineering.

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EXPERIENCE

AI PRODUCT ENGINEER | Independent Product Development

Jun '25 - Present

Deliberately pursued end-to-end AI product ownership, progressing from focused applications to deployed products solving real user problems.

[CineMind](#) — AI-Native Movie Discovery Engine

Solo-built and deployed a 120K-film movie-discovery platform supporting natural-language search, multi-movie similarity search, and exploratory browsing.

- Architected hybrid retrieval across 8 Qdrant vector spaces and PostgreSQL lexical/metadata indexes, decomposing freeform intent into targeted searches, deterministically fusing results, and supporting multi-signal similarity matching from one or more seed movies
- Built reproducible evaluation systems for subjective LLM and retrieval quality, combining rubric-based, bucket-stratified, contamination-controlled LLM-as-judge evaluations with precision, recall, MRR, and NDCG regression testing to guide model, prompt, schema, and architecture decisions.
- Engineered a crash-safe, idempotent ingestion pipeline that narrowed ~1M raw titles to ~120K searchable films through multi-source data collection, weighted quality scoring, survival-curve derivative analysis, and staged generation of LLM-derived metadata.
- Reduced Qdrant retrieval latency by 10x through profiling and query tuning, while cutting LLM data-generation costs 4x through Batch API processing, context distillation, schema slimming, and task-specific model routing.
- Shipped the full production stack: Next.js/React frontend with streamed pipeline progress, FastAPI APIs, PostgreSQL, Qdrant, and Redis in Docker Compose on AWS EC2, protected by Cloudflare rate limiting, structured-output guardrails, and production observability.

Additional AI systems

- Developed a provider-agnostic memory system for coding agents that hierarchically loads project context and promotes session learnings into human-reviewed architecture decisions, preventing contradictory rework across sessions without bloating prompts.
- Built [InterviewPro](#), a multi-agent interview-prep platform with parallel web research, schema-validated outputs, prompt-injection guardrails, and a six-judge evaluation panel

SOFTWARE ENGINEER | Meta (Instagram)

Jun '22 - May '25

Shipped across client UI, backend services, and data analytics through A/B experiments for millions of users. Promoted to mid-level at 1.5 years.

- Designed and built a new data instrumentation system fixing self-identified reliability gaps that had caused a 34% regression in engagement metrics; produced insights that initiated data-driven product iteration and eventually adopted as the org's gold standard.
- Owned Creator Goals features end-to-end in an unfamiliar stack amid shifting designs. Built the full data collection system for experiment launch and produced user-behavior insights that became the team's primary deliverable and shaped future roadmaps
- Expanded Creator Achievements from Reels to posts and Stories by reverse-engineering undocumented, high-traffic creation flows and generalizing fragile code across legacy media types without regressions while building sequential milestones in parallel.

SOFTWARE ENGINEER INTERN, iOS | The Home Depot, OrangeWorks Innovation Labs

Aug '20 - Dec '20

- Led a zero-to-one Google Design Sprint, using weeks of stakeholder interviews to identify in-store wait times as a key pain point, then independently designed and built an iOS prototype for pre-authorized Pro purchasing allowances.
- Pitched the prototype to executives and got approval for production development; later [launched into Home Depot's Pro Xtra program](#)

SOFTWARE ENGINEER INTERN, iOS | MealMe (now Satsuma.ai)

Jun '20 - Aug '20

- At a four-person startup, independently replaced Google Maps with an Apple MapKit/Yelp data pipeline, eliminating per-user API costs that had blocked paid acquisition and product expansion
- Conducted competitor research and user interviews to redesign the iOS browsing experience, doubling daily active users at launch

SKILLS

- **AI/LLM:** RAG, agents, LLM evaluation, semantic/vector search, tool calling, structured outputs, guardrails, OpenAI, Anthropic, Gemini
- **Application & Data:** Python, TypeScript, SQL, Swift; FastAPI, React, Next.js, Pydantic, PostgreSQL, Qdrant, Redis, GraphQL
- **Infrastructure & AI Development:** AWS, Docker; Claude Code, Codex, Cursor, prompt/context engineering, agentic development

EDUCATION

Georgia Institute of Technology — B.S. in Computer Science, Highest Honors

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